

Chapter 3: Linear combinations, span and subspaces

Section 3.1: Linear combinations

In short, forming linear combinations is a way of producing new vectors from old.

Suppose you have some vectors $v_1, v_2, \dots, v_n$ of a vector space. A *linear combination* of these is any vector that can be obtained from the v_j by “multiplication by a scalar” and “addition of vectors”.

That is, v is a linear combination of $v_1, v_2, \dots, v_n$ if there are scalars (=real numbers) $\lambda_1, \lambda_2, \dots, \lambda_n$ such that

$$v = \lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n .$$

1 Linear combinations I

V an $\mathbb{R}$ -vector space, $v_1, v_2, \dots, v_n \in V$

Definition. A *linear combination* of the vectors v_j is a vector of the form

$$\lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n , \quad \text{also written as} \quad \sum_{j=1}^n \lambda_j v_j ,$$

where the λ_j are scalars (*i.e.*, real numbers).

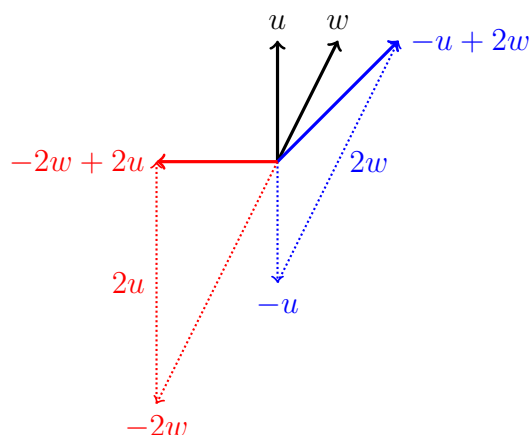
Example: The vector $\begin{pmatrix} 2 \\ 9 \end{pmatrix} \in \mathbb{R}^2$ is a linear combination of the vectors $\begin{pmatrix} 1 \\ -3 \end{pmatrix}$ and $\begin{pmatrix} 2 \\ -1 \end{pmatrix}$ since

$$\begin{pmatrix} 2 \\ 9 \end{pmatrix} = -4 \begin{pmatrix} 1 \\ -3 \end{pmatrix} + 3 \begin{pmatrix} 2 \\ -1 \end{pmatrix} .$$

2 Linear combinations in $\mathbb{R}^2$: A picture

With $u = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$ and $w = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$, we form the linear combinations

$$\begin{pmatrix} -2 \\ 0 \end{pmatrix} = -2w + 2u \text{ and } \begin{pmatrix} 2 \\ 2 \end{pmatrix} = -u + 2w :$$



3 Linear combinations: Example 1

Recall: $\mathcal{P}_n$ = the space of polynomials of degree $\leq n$ (slides 22–23 of Chapter 2)

Claim: Every element of $\mathcal{P}_n$ is a linear combination of these polynomials:

$$x^n - x^{n-1}, \quad x^{n-1} - x^{n-2}, \quad \dots, \quad x^2 - x, \quad x - 1, \quad 1.$$

Proof. For $a_n x^n + a_{n-1} x^{n-1} + \dots + a_1 x + a_0 \in \mathcal{P}_n$ we have

$$\begin{aligned} & a_n x^n + a_{n-1} x^{n-1} + \dots \\ & \dots + a_1 x + a_0 = a_n \cdot (x^n - x^{n-1}) \\ & \quad + (a_n + a_{n-1}) \cdot (x^{n-1} - x^{n-2}) \\ & \quad + (a_n + a_{n-1} + a_{n-2}) \cdot (x^{n-2} - x^{n-3}) \\ & \quad + \dots \\ & \quad + (a_n + a_{n-1} + \dots + a_1) \cdot (x - 1) \\ & \quad + (a_n + a_{n-1} + \dots + a_1 + a_0) \cdot 1 \end{aligned}$$

■

4 Linear combinations: Example 2

Suppose we are given the following vectors in $V = \mathbb{R}^3$:

$$v_1 = \begin{pmatrix} 2 \\ 2 \\ 6 \end{pmatrix} \quad v_2 = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} \quad v_3 = \begin{pmatrix} 3 \\ 1 \\ 13 \end{pmatrix}$$

Question: Can we find out if it is possible to write $w = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}$ as a linear combination of v_1 , v_2 and v_3 ?

Answer: Write w as a linear combination with unknown coefficients:

$$\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = w \stackrel{\text{want}}{=} \lambda_1 v_1 + \lambda_2 v_2 + \lambda_3 v_3 = \begin{pmatrix} 2\lambda_1 + \lambda_2 + 3\lambda_3 \\ 2\lambda_1 + \lambda_2 + \lambda_3 \\ 6\lambda_1 + 3\lambda_2 + 13\lambda_3 \end{pmatrix} \quad (\dagger)$$

The question now becomes: *Can we find suitable scalars λ_j ?*

5 Linear combinations: Example 2 (continued)

The vector equation $(\dagger)$ can be re-written as a system of linear equations:

$$\left. \begin{array}{rrrr} 2\lambda_1 & + & \lambda_2 & + & 3\lambda_3 & = & 1 \\ 2\lambda_1 & + & \lambda_2 & + & \lambda_3 & = & 2 \\ 6\lambda_1 & + & 3\lambda_2 & + & 13\lambda_3 & = & 1 \end{array} \right\} \quad (\ddagger)$$

(Note: λ_1 , λ_2 and λ_3 are the variables!) The question “can we find suitable scalars λ_j ?” now translates into the question:

Does the system $(\ddagger)$ have a solution?

It does not matter *how many* solutions there are in this case, or *what* the solutions are, **as long as at least one solution exists.**

6 Linear combinations: Example 2 (continued)

Solve system (‡); using Gauss elimination in matrix notation:

$$\begin{pmatrix} 2 & 1 & 3 & | & 1 \\ 2 & 1 & 1 & | & 2 \\ 6 & 3 & 13 & | & 1 \end{pmatrix} \xrightarrow[\substack{(II)-(I) \\ (III)-3 \cdot (I)}]{\substack{(II)-(I) \\ (III)-3 \cdot (I)}} \begin{pmatrix} 2 & 1 & 3 & | & 1 \\ 0 & 0 & -2 & | & 1 \\ 0 & 0 & 4 & | & -2 \end{pmatrix} \xrightarrow{(III)+2 \cdot (II)} \begin{pmatrix} 2 & 1 & 3 & | & 1 \\ 0 & 0 & -2 & | & 1 \\ 0 & 0 & 0 & | & 0 \end{pmatrix}$$

We see that *this system has a solution* (at least one), hence:

Answer: w is indeed a linear combination of v_1 , v_2 and v_3 .

We can get an *explicit* expression for w : *choose* a value for the parameter, and *calculate* the other variables. For example, *choose* $\lambda_2 = 1/2$ and *calculate* $\lambda_3 = -1/2$ and $\lambda_1 = 1$, so that

$$\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} = w = v_1 + \frac{1}{2}v_2 - \frac{1}{2}v_3 = \begin{pmatrix} 2 \\ 2 \\ 6 \end{pmatrix} + \frac{1}{2} \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} - \frac{1}{2} \begin{pmatrix} 3 \\ 1 \\ 13 \end{pmatrix} .$$

7 Linear combinations II

One can also consider linear combinations of vectors taken from a possibly infinite set, but note:

A linear combination is always a **finite** sum of scalar multiples of vectors, and *not* an infinite sum.

Let V an $\mathbb{R}$ -vector space, and let $S \subseteq V$ be a non-empty subset.

Definition. A *linear combination of vectors in S* is a vector of the form

$$\lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_\ell v_\ell , \quad \text{also written} \quad \sum_{j=1}^{\ell} \lambda_j v_j ,$$

where $\ell \in \mathbb{N}$, the v_j are elements of S , and the λ_j are scalars (*i.e.*, real numbers).

If $S = \{v_1, v_2, \dots, v_n\}$ is a finite set then this is the same as linear combinations of the vectors $v_1, \dots, v_n$.

Section 3.2: The span of a set

8 The span of a set

Definition. Let V be a vector space over $\mathbb{R}$ and let $S \subseteq V$ be a non-empty subset of V . The *span of S* , denoted $\text{span}(S)$ or $\langle S \rangle$, is the set of all linear combinations of vectors in S :

$$\text{span}(S) = \langle S \rangle = \left\{ \sum_{j=1}^{\ell} \lambda_j v_j \mid \ell \in \mathbb{N}, \text{ all } \lambda_j \in \mathbb{R}, \text{ all } v_j \in S \right\}.$$

We define $\text{span}(\emptyset) = \{\mathbf{0}\}$ (as a useful convention).

So $\text{span}(S)$ is defined for all subsets $S \subseteq V$. We have inclusions:

$$S \subseteq \text{span}(S) \subseteq V$$

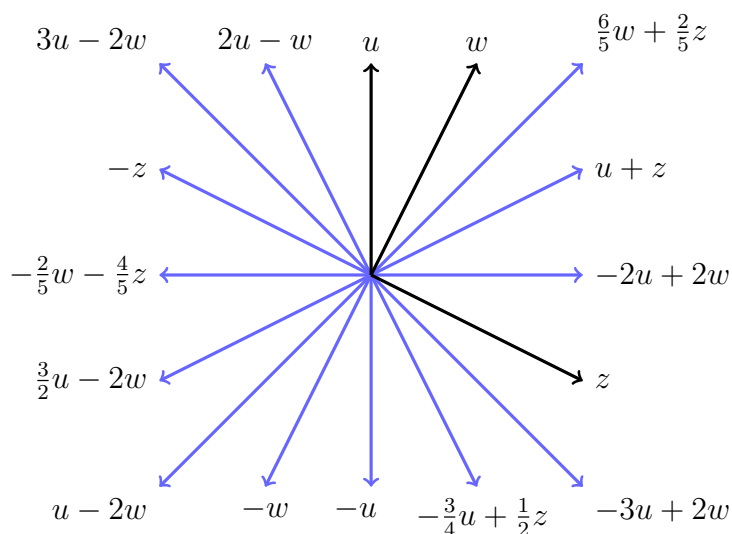
(For all $s \in S$ have $s = 1s \in \text{span}(S)$, hence $S \subseteq \text{span}(S)$).

An element of $\text{span}(S)$ is a linear combination of vectors in S , hence is a vector in V . Thus $\text{span}(S) \subseteq V$ holds.)

9 Span: a picture

$$u = \begin{pmatrix} 0 \\ 2 \end{pmatrix} \quad w = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad z = \begin{pmatrix} 2 \\ -1 \end{pmatrix} \quad \text{span}\{u, w, z\} = \mathbb{R}^2$$

Intuitively: we can form linear combinations of u , w and z to result in vectors “pointing in any direction”, as indicated by sketch below.



10 Example for “span”: The unit vectors span $\mathbb{R}^n$

In the $\mathbb{R}$ -vector space $\mathbb{R}^n$ consider the “unit vectors” e_j , $1 \leq j \leq n$, where e_j has j th entry 1, and all other entries 0:

$$e_1 = \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad e_2 = \begin{pmatrix} 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \dots, \quad e_n = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix}.$$

With this notation we have:

$$\text{span}(\{e_1, e_2, \dots, e_n\}) = \mathbb{R}^n$$

In words: *The unit vectors $e_1, e_2, \dots, e_n \in \mathbb{R}^n$ span the vector space $\mathbb{R}^n$.*

11 Example for “span” (continued)

Proof. Recall $\text{span}(\{e_1, e_2, \dots, e_n\}) = \{\sum_{j=1}^n \lambda_j e_j \mid \text{all } \lambda_j \in \mathbb{R}\}$.

We know $\text{span}(\{e_1, e_2, \dots, e_n\}) \subseteq \mathbb{R}^n$ (by definition of “span”).

Need to prove $\text{span}(\{e_1, e_2, \dots, e_n\}) \supseteq \mathbb{R}^n$. So let $x \in \mathbb{R}^n$, where x has components $x_1, x_2, \dots, x_n$ as usual. Then

$$\begin{aligned} x = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} &= x_1 \begin{pmatrix} 1 \\ 0 \\ \vdots \\ 0 \end{pmatrix} + x_2 \begin{pmatrix} 0 \\ 1 \\ \vdots \\ 0 \end{pmatrix} + \dots + x_n \begin{pmatrix} 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix} \\ &= \sum_{j=1}^n x_j e_j \in \text{span}(\{e_1, e_2, \dots, e_n\}) . \quad \blacksquare \end{aligned}$$

12 Another example for “span”

Recall: $\mathcal{P}_n$ = the space of polynomials of degree $\leq n$ (slides 22–23 of Chapter 2).

Claim: *The vector space $\mathcal{P}_n$ is spanned by the $n+1$ vectors*

$$x^n - x^{n-1} , \quad x^{n-1} - x^{n-2} , \quad \dots , \quad x^2 - x , \quad x - 1 , \quad 1 .$$

Proof. The span of these vectors is a subset of $\mathcal{P}_n$, by definition of “span”.

We need to show that conversely $\mathcal{P}_n$ is a subset of the span. That is: we need to show that *every element of $\mathcal{P}_n$ is a linear combination of the vectors listed above.*

This was already checked (slide 3 of this Chapter), so we are done! ■

13 Describing the span explicitly: Example 1

Suppose we are given the following vectors in $V = \mathbb{R}^3$:

$$v_1 = \begin{pmatrix} 2 \\ 2 \\ 6 \end{pmatrix} \quad v_2 = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix} \quad v_3 = \begin{pmatrix} 3 \\ 1 \\ 13 \end{pmatrix}$$

Question: How can we find out explicitly which vectors $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$ are in $\text{span}(\{v_1, v_2, v_3\})$?

One answer is given by the definition of the “span”:

$$\text{span}(\{v_1, v_2, v_3\}) = \{\lambda_1 v_1 + \lambda_2 v_2 + \lambda_3 v_3 \mid \lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}\}$$

So v is in $\text{span}(\{v_1, v_2, v_3\})$ if and only if there are scalars $\lambda_1, \lambda_2, \lambda_3$ with $v = \lambda_1 v_1 + \lambda_2 v_2 + \lambda_3 v_3$.

But we can give a different description which is often more convenient.

14 Describing the span explicitly: Example 1 (continued 1)

Again: The vector $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$ lies in $\text{span}(\{v_1, v_2, v_3\})$ if and only if there are $\lambda_1, \lambda_2, \lambda_3 \in \mathbb{R}$ so that

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda_1 v_1 + \lambda_2 v_2 + \lambda_3 v_3 = \begin{pmatrix} 2\lambda_1 + \lambda_2 + 3\lambda_3 \\ 2\lambda_1 + \lambda_2 + \lambda_3 \\ 6\lambda_1 + 3\lambda_2 + 13\lambda_3 \end{pmatrix}.$$

This happens if and only if the following system of linear equations **has a solution**:

$$\left. \begin{array}{rrrr} 2\lambda_1 & + & \lambda_2 & + & 3\lambda_3 & = & x \\ 2\lambda_1 & + & \lambda_2 & + & \lambda_3 & = & y \\ 6\lambda_1 & + & 3\lambda_2 & + & 13\lambda_3 & = & z \end{array} \right\} \quad (\bowtie)$$

(Note: λ_1, λ_2 and λ_3 are the variables!) We don't need to find a solution. We only need to find out if a solution exists (or not).

15 Describing the span explicitly: Example 1 (continued 2)

Solve system $(\bowtie)$; using Gauss elimination in matrix notation:

$$\begin{aligned} \left(\begin{array}{ccc|c} 2 & 1 & 3 & x \\ 2 & 1 & 1 & y \\ 6 & 3 & 13 & z \end{array} \right) & \xrightarrow[\text{(III)-3}\cdot\text{(I)}]{\text{(II)}-\text{(I)}} \left(\begin{array}{ccc|c} 2 & 1 & 3 & x \\ 0 & 0 & -2 & y-x \\ 0 & 0 & 4 & z-3x \end{array} \right) \\ & \xrightarrow{\text{(III)}+2\cdot\text{(II)}} \left(\begin{array}{ccc|c} 2 & 1 & 3 & x \\ 0 & 0 & -2 & y-x \\ 0 & 0 & 0 & z+2y-5x \end{array} \right) \end{aligned}$$

We can now read off the required condition from the last row:

The vector $\begin{pmatrix} x \\ y \\ z \end{pmatrix}$ lies in $\text{span}(\{v_1, v_2, v_3\})$ if and only if the equation $z + 2y - 5x = 0$ is satisfied.

In symbols:

$$\text{span}(\{v_1, v_2, v_3\}) = \left\{ \begin{pmatrix} x \\ y \\ 5x - 2y \end{pmatrix} \mid x, y \in \mathbb{R} \right\}$$

16 Describing the span explicitly: Example 1 (continued 3)

Question: Given that $u = \begin{pmatrix} x \\ y \\ z \end{pmatrix} \in \text{span}(\{v_1, v_2, v_3\})$, how can we write u as a linear combination of v_1 , v_2 and v_3 ?

We have solved the relevant system ($\boxtimes$) already. We choose $\lambda_2 = 0$ (to make life easy); recall that λ_2 is a parameter. Next, we calculate

$$\lambda_3 = \frac{x - y}{2} \quad \text{and} \quad \lambda_1 = \frac{x - 3\lambda_3}{2} = \frac{3y - x}{4}.$$

That is, we find

$$u = \begin{pmatrix} x \\ y \\ z \end{pmatrix} = \lambda_1 v_1 + \lambda_3 v_3 = \frac{3y - x}{4} \cdot \begin{pmatrix} 2 \\ 2 \\ 6 \end{pmatrix} + \frac{x - y}{2} \cdot \begin{pmatrix} 3 \\ 1 \\ 13 \end{pmatrix}.$$

Recall $z = 5x - 2y$ as $u \in \text{span}(\{v_1, v_2, v_3\})$.

We could also choose $\lambda_2 = 1.572$ if we wanted to.

17 Describing the span explicitly: Example 2

$v_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ and $v_2 = \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}$ vectors in $\mathbb{R}^3$

Question: Can we describe $\text{span}\{v_1, v_2\}$ explicitly?

$$\begin{aligned}\text{span}\{v_1, v_2\} &= \left\{ \lambda \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \mid \lambda, \mu \in \mathbb{R} \right\} \\ &= \left\{ \begin{pmatrix} \lambda + \mu \\ 2\mu \\ \lambda - \mu \end{pmatrix} \mid \lambda, \mu \in \mathbb{R} \right\}\end{aligned}$$

So $b = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} \in \text{span}\{v_1, v_2\}$ if and only if b can be written as

$$\begin{pmatrix} \lambda + \mu \\ 2\mu \\ \lambda - \mu \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}, \text{ for some } \lambda, \mu \in \mathbb{R}.$$

18 Describing the span explicitly: Example 2 (continued)

This is equivalent to the following system of linear equations:

$$\left. \begin{array}{rcl} \lambda & + & \mu & = & b_1 \\ & & 2\mu & = & b_2 \\ \lambda & - & \mu & = & b_3 \end{array} \right\}$$

We bring this into row echelon form (exercise!) and obtain

$$\left. \begin{array}{rcl} \lambda & + & \mu & = & b_1 \\ & & 2\mu & = & b_2 \\ & & 0 & = & b_1 - b_2 - b_3 \end{array} \right\}$$

This has no solution if $b_3 \neq b_1 - b_2$, and has precisely one solution otherwise. Hence:

$$\text{span}\{v_1, v_2\} = \left\{ \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix} \mid b_3 = b_1 - b_2 \right\}$$

19 Monotonicity of $\text{span}(\cdot)$

The following property of “span” is of fundamental importance: Applying “span” to larger sets results in larger subspaces.

Let V be an $\mathbb{R}$ -vector space. For subsets T and S of V , if $T \subseteq S$ then $\text{span}(T) \subseteq \text{span}(S)$.

Proof. An element v of $\text{span}(T)$ is, by definition, a linear combination of vectors in T . But T is a subset of S , so every vector in T is a vector in S and v is thus a linear combination of vectors in S . This means that $v \in \text{span}(S)$.

So every element of $\text{span}(T)$ is also an element of $\text{span}(S)$, which means precisely $\text{span}(T) \subseteq \text{span}(S)$. ■

Informally: if you make the set S larger then the set $\text{span}(S)$ becomes larger too.

A sub/space/ of a vector space V is a sub/set/ W of V that is itself a vector space, with addition and multiplication by scalar inherited from V .

We will give a formal definition of subspaces, provide a criterion that allows us to recognise when a given sub/set/ is a sub/space/, and verify that the span (of any set of vectors) is in fact a subspace of the original vector space.

Section 3.3: Subspaces

20 Subspaces

Definition. Let V be a vector space over $\mathbb{R}$. A subset $W \subseteq V$ is called a *subspace* of V if

- **(Sub 0)** $\underline{0} \in W$ (so in particular $W \neq \emptyset$);
- **(Sub 1)** $u + v \in W$ whenever $u, v \in W$;
- **(Sub 2)** $\lambda w \in W$ whenever $w \in W$ and $\lambda \in \mathbb{R}$.

A subspace W , equipped with “addition of vectors” and “multiplication by scalar” operations inherited from the ambient space V , is a vector space in its own right.

The eight conditions (A1)–(S4) for a vector space are satisfied for W since they hold in V .

21 Subspaces: first examples

Observation: Every vector space V has two subspaces,

- the space $\{\underline{0}\} \subseteq V$,
- the space V itself.

More interesting is the following result:

The set of solutions of a homogeneous system of linear equations in n variables is a subspace of $\mathbb{R}^n$.

In fact, on slides 16–17 of Chapter 2 we proved this already (but without using the word “subspace”).

Remark: The Theorem only holds for *homogeneous* systems (all right-hand sides 0).

22 Subspaces: Another example

Recall the vector space $\mathcal{S} = \{(a_n)_{n=1}^\infty \mid \text{all } a_n \in \mathbb{R}\}$ of sequences of real numbers from slide 18 of Chapter 2. Define

$$\mathcal{S}_{\text{conv}} = \{(a_n)_{n=1}^\infty \in \mathcal{S} \mid \text{the sequence } (a_n)_{n=1}^\infty \text{ converges}\} \subset \mathcal{S}$$

The set $\mathcal{S}_{\text{conv}}$ is a subspace of $\mathcal{S}$.

Proof. Need to check three conditions (see slide 20 of this Chapter):

- **(Sub 0)** $\underline{0} = (0)_{n=1}^\infty$ converges, so $\underline{0} \in \mathcal{S}_{\text{conv}}$

- **(Sub 1)** If $a = (a_n)_{n=1}^\infty$ and $b = (b_n)_{n=1}^\infty$ are in $\mathcal{S}_{\text{conv}}$ they converge; but then so does $a + b = (a_n + b_n)_{n=1}^\infty$, hence $a + b \in \mathcal{S}_{\text{conv}}$
- **(Sub 2)** If $a = (a_n)_{n=1}^\infty$ then a converges; but then so does $\lambda a = (\lambda a_n)_{n=1}^\infty$ for any $\lambda \in \mathbb{R}$ so that $\lambda a \in \mathcal{S}_{\text{conv}}$. For (Sub 1) and (Sub 2) we used the “algebra of limits”.

■

23 Detecting subspaces: The subspace criterion

Subspace criterion

Let V be an $\mathbb{R}$ -vector space and let $W \subseteq V$ be a non-empty subset. The following statements are equivalent:

- **(i)** W is a subspace of V
- **(ii)** $\lambda v + \mu w \in W$ for all $v, w \in W$ and all $\lambda, \mu \in \mathbb{R}$
- **(iii)** $\lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_n v_n \in W$ for all $n \geq 1$, all $v_1, \dots, v_n \in W$ and all $\lambda_1, \dots, \lambda_n \in \mathbb{R}$
- **(iv)** $\text{span}(W) \subseteq W$

Part (ii) is a convenient criterion to decide whether a given *non-empty* subset W of V is a subspace or not.

We prove (i) $\Leftrightarrow$ (ii) only, leaving (ii) $\Leftrightarrow$ (iii) as an exercise.

Part (iv) is just a reformulation of (iii).

24 Detecting subspaces: Proof of (i) $\Rightarrow$ (ii)

Suppose (i) holds: W is a subspace of V . Then W satisfies the conditions (Sub 0), (Sub 1) and (Sub 2) from slide 20 of this Chapter.

Let $v, w \in W$ and $\lambda, \mu \in \mathbb{R}$ be given. We need to show that $\lambda v + \mu w \in W$. Here's the argument:

- $\lambda v \in W$, by (Sub 2);
- $\mu w \in W$, by (Sub 2);
- and thus $(\lambda v) + (\mu w) \in W$, by (Sub 1).

This shows that the implication (i) $\Rightarrow$ (ii) holds.

25 Detecting subspaces: Proof of (ii) $\Rightarrow$ (i)

Suppose (ii) holds: $\lambda v + \mu w \in W$ for all $v, w \in W$ and all $\lambda, \mu \in \mathbb{R}$.

Need to show: W satisfies (Sub 0), (Sub 1) and (Sub 2).

- **(Sub 0)** Since $W \neq \emptyset$ we can choose $w \in W$. Then

$$\underline{0} = w - w = 1 \cdot w + (-1) \cdot w \in W \quad \text{by (ii) .}$$

- **(Sub 1)** Let $u, v \in W$. Then $u + v = 1 \cdot u + 1 \cdot v \in W$ by (ii).
- **(Sub 2)** Let $w \in W$ and $\lambda \in \mathbb{R}$. Then

$$\lambda w = \lambda w + \underline{0} = \lambda w + 0w \in W \quad \text{by (ii) .}$$

This shows that the implication (ii) $\Rightarrow$ (i) holds. ■

26 Example for detecting subspaces

In $\mathcal{M}_{3,3}$, the vector space of 3×3 -matrices, we define a subset

$$\begin{aligned} \mathcal{U}_3 &= \text{set of upper triangular } 3 \times 3\text{-matrices} \\ &= \left\{ \begin{pmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ a_{2,1} & a_{2,2} & a_{2,3} \\ a_{3,1} & a_{3,2} & a_{3,3} \end{pmatrix} \left| \begin{array}{l} a_{2,1} = 0 \\ a_{3,1} = 0 \\ a_{3,2} = 0 \end{array} \right. \right\} \\ &= \left\{ \begin{pmatrix} a_{1,1} & a_{1,2} & a_{1,3} \\ 0 & a_{2,2} & a_{2,3} \\ 0 & 0 & a_{3,3} \end{pmatrix} \left| a_{ij} \in \mathbb{R} \text{ for } 1 \leq i \leq j \leq 3 \right. \right\} \end{aligned}$$

The set $\mathcal{U}_3$ is a subspace of $\mathcal{M}_{3,3}$.

Clearly $\mathcal{U}_3 \neq \emptyset$ since the zero matrix is an element of $\mathcal{U}_3$.

27 Example for detecting subspaces (continued)

For all $A = (a_{i,j}) \in \mathcal{U}_3$, $B = (b_{i,j}) \in \mathcal{U}_3$ and $\lambda, \mu \in \mathbb{R}$ we have

$$\lambda A + \mu B = \begin{pmatrix} \lambda a_{1,1} + \mu b_{1,1} & \lambda a_{1,2} + \mu b_{1,2} & \lambda a_{1,3} + \mu b_{1,3} \\ \lambda a_{2,1} + \mu b_{2,1} & \lambda a_{2,2} + \mu b_{2,2} & \lambda a_{2,3} + \mu b_{2,3} \\ \lambda a_{3,1} + \mu b_{3,1} & \lambda a_{3,2} + \mu b_{3,2} & \lambda a_{3,3} + \mu b_{3,3} \end{pmatrix}$$

$$\left. \begin{array}{l} A \in \mathcal{U}_3 \Rightarrow a_{2,1} = 0 \\ B \in \mathcal{U}_3 \Rightarrow b_{2,1} = 0 \end{array} \right\} \Rightarrow \lambda a_{2,1} + \mu b_{2,1} = 0$$

Similarly $\lambda a_{3,1} + \mu b_{3,1} = 0 = \lambda a_{3,2} + \mu b_{3,2}$.

Hence $\lambda A + \mu B \in \mathcal{U}_3$.

From subspace criterion (slide 23 of this Chapter) we get that $\mathcal{U}_3$ is a subspace of $\mathcal{M}_{3,3}$ as claimed. ■

28 The span is a subspace

Let V be a vector space and $S \subseteq V$ be a subset. Then $\text{span}(S)$ is a subspace of V .

Proof. We use the subspace criterion from slide 23 of this Chapter.

We have $\text{span}(S) \subseteq V$. Also, $\text{span}(S) \neq \emptyset$:

- if $S = \emptyset$ then $\text{span}(S) = \text{span}(\emptyset) = \{\underline{0}\} \neq \emptyset$ by convention;
- if $S \neq \emptyset$ then $\text{span}(S) \neq \emptyset$ since $S \subseteq \text{span}(S)$.

29 The span is a subspace (continued)

Take elements

$$\begin{aligned}u &= \lambda_1 u_1 + \cdots + \lambda_n u_n \in \text{span}(S) \\v &= \mu_1 v_1 + \cdots + \mu_m v_m \in \text{span}(S)\end{aligned}$$

where $u_1, \dots, u_n, v_1, \dots, v_m$ are elements of S and $\lambda_1, \dots, \lambda_n, \mu_1, \dots, \mu_m$ are scalars. For every $\lambda, \mu \in \mathbb{R}$,

$$\begin{aligned}\lambda u + \mu v &= \lambda(\lambda_1 u_1 + \cdots + \lambda_n u_n) + \mu(\mu_1 v_1 + \cdots + \mu_m v_m) \\&= (\lambda \lambda_1) u_1 + \cdots + (\lambda \lambda_n) u_n + (\mu \mu_1) v_1 + \cdots + (\mu \mu_m) v_m\end{aligned}$$

which is in $\text{span}(S)$. Thus, $\text{span}(S)$ is a subspace. ■

30 $\text{span}(S)$ is the smallest subspace containing S

V a vector space over $\mathbb{R}$, and $S \subseteq V$ a non-empty subset

$\text{span}(S)$ is called the vector space *spanned* or *generated* by S , and is sometimes denoted $\langle S \rangle$.

Let V be a vector space over $\mathbb{R}$ and $S \subseteq V$ be a subset. Then the following statements hold:

- $S \subseteq \text{span}(S)$
- if U is any subspace of V with $S \subseteq U$, then $\text{span}(S) \subseteq U$

That is, $\text{span}(S)$ is the smallest subspace of V containing S .

Proof. $S \subseteq \text{span}(S)$ was observed before.

Next, if U contains S then U contains all linear combinations of elements of S , by the subspace criterion (slide 23 of this Chapter). Hence U contains the set of all these linear combinations, which is precisely $\text{span}(S)$. ■

Section 3.4: Finitely generated spaces, redundant generators

We are mainly interested in vector spaces that can be described as the span of *finitely many vectors*; for example, $\mathbb{R}^n$ is of this type (slide 10 of this chapter). Most of the vector spaces we are interested in fall into this category. These vector spaces are called “finitely generated”.

31 Finite generation

Definition. An $\mathbb{R}$ -vector space V is called *finitely generated* if there is a *finite* set $S = \{v_1, v_2, \dots, v_\ell\} \subseteq V$ with $\text{span}(S) = V$.

(Recall: $\text{span}(S)$ is the set of linear combinations of the vectors $v_1, \dots, v_\ell$.)

Explicitly: the vector space V is finitely generated if and only if there are vectors $v_1, v_2, \dots, v_\ell \in V$ with

$$V = \left\{ \lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_\ell v_\ell \mid \lambda_1, \lambda_2, \dots, \lambda_\ell \in \mathbb{R} \right\}.$$

This means that *every* vector of V is a linear combination of the v_j .

If $S \subseteq V$ is finite, then $U = \text{span}(S)$ is a finitely generated vector space, even if V is not.

32 Examples for finite generation

$\mathbb{R}^n$ is finitely generated.

We saw that on slide 10 of this chapter:

$$\text{span}(\{e_1, e_2, \dots, e_n\}) = \mathbb{R}^n$$

(Here $e_1, e_2, \dots, e_n \in \mathbb{R}^n$ are the unit vectors, as usual.)

$\mathcal{P}_n$ is finitely generated.

We saw that on slide 12:

$$\mathcal{P}_n = \text{span} \left\{ x^n - x^{n-1}, x^{n-1} - x^{n-2}, \dots, x^2 - x, x - 1, 1 \right\}$$

33 Examples for finite generation (continued 1)

Claim: $M = \left\{ \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n \mid x_1 + x_2 + \dots + x_n = 0 \right\}$ is a finitely generated subspace of $\mathbb{R}^n$.

Proof. M is a subspace of $\mathbb{R}^n$ as it is the set of solutions of a “homogeneous system of linear equations” (with a single equation).

Now consider the vectors

$$v_1 = \begin{pmatrix} 1 \\ -1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad v_2 = \begin{pmatrix} 1 \\ 0 \\ -1 \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad \dots, \quad v_{n-1} = \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \\ -1 \end{pmatrix} \in \mathbb{R}^n.$$

Claim: $M = \text{span} \{v_1, v_2, \dots, v_{n-1}\}$. (This shows that M is finitely generated.)

34 Examples for finite generation (continued 2)

Proof of claim: We have $v_i \in M$ for each $i = 1, \dots, n-1$, so

$$\{v_1, v_2, \dots, v_{n-1}\} \subseteq M \text{ and thus } \text{span} \{v_1, v_2, \dots, v_{n-1}\} \subseteq M$$

since $\text{span} \{v_1, v_2, \dots, v_{n-1}\}$ is the smallest subspace of V containing the set $\{v_1, v_2, \dots, v_{n-1}\}$, see slide 30 of this chapter.

Still need to prove: $\text{span} (\{v_1, v_2, \dots, v_{n-1}\}) \supseteq M$.

In words: want to show that *every vector* $b \in M$ is a linear combination of the vectors $v_1, v_2, \dots, v_{n-1}$.

Take $b = \begin{pmatrix} b_0 \\ b_1 \\ \vdots \\ b_{n-1} \end{pmatrix} \in M$. Being in M , its components satisfy $b_0 + b_1 + \dots + b_{n-1} = 0$, that is, $b_0 = -(b_1 + b_2 + \dots + b_{n-1})$.

Want to find scalars $\lambda_1, \dots, \lambda_{n-1}$ with

$$\lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_{n-1} v_{n-1} = b .$$

35 Examples for finite generation (continued 3)

The equation $\lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_{n-1} v_{n-1} = b$ is equivalent to the following system of linear equations:

$$\left. \begin{array}{ccccccc} \lambda_1 & + & \lambda_2 & + & \cdots & + & \lambda_{n-1} & = & -(b_1 + b_2 + \dots + b_{n-1}) \\ -\lambda_1 & & & & & & & = & b_1 \\ & & -\lambda_2 & & & & & = & b_2 \\ & & & & \ddots & & & \vdots & \\ & & & & & & -\lambda_{n-1} & = & b_{n-1} \end{array} \right\}$$

A solution is given by $\lambda_i = -b_i$, for $1 \leq i \leq n-1$.

This means that

$$b = (-b_1)v_1 + (-b_2)v_2 + \dots + (-b_{n-1})v_{n-1} \in \text{span}(\{v_1, v_2, \dots, v_{n-1}\}) ,$$

as required. ■

Omitting redundant generators: idea

We have seen that some vector spaces are “generated” (spanned) by finitely many vectors. It is interesting, both for computations and for theory, to try to get away with as few vectors as possible.

Roughly speaking, $\text{span}(S)$ does not change if we make S smaller *provided* the vectors we leave out are in the span of the remaining vectors. This is formalised in the process of “omitting redundant generators”.

36 Omitting redundant generators

Let S be a subset of the vector space V , and let $T \subseteq S$.

Suppose that every vector $t \in T$ is in $\text{span}(S \setminus T)$, *i.e.*, suppose that $T \subseteq \text{span}(S \setminus T)$.

Then $\text{span}(S \setminus T) = \text{span}(S)$.

Proof. “ $\subseteq$ ” We have $S \setminus T \subseteq S$ and hence $\text{span}(S \setminus T) \subseteq \text{span}(S)$ by monotonicity of span , see slide 19.

“ $\supseteq$ ” We have $S \setminus T \subseteq \text{span}(S \setminus T)$, but also $T \subseteq \text{span}(S \setminus T)$ by hypothesis. This means that

$$S = (S \setminus T) \cup T \subseteq \text{span}(S \setminus T) .$$

So $\text{span}(S \setminus T)$ is a subspace containing S .

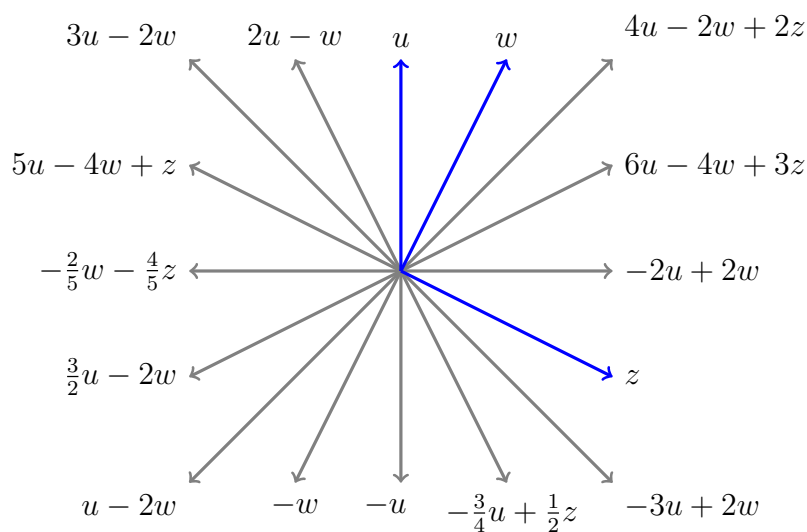
But $\text{span}(S)$ is the *smallest* subspace of V containing S , cf. slide 30, so that $\text{span}(S) \subseteq \text{span}(S \setminus T)$ as required. ■

37 Omitting redundant generators: picture 1 of 4

$$u = \begin{pmatrix} 0 \\ 2 \end{pmatrix} \quad w = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad z = \begin{pmatrix} 2 \\ -1 \end{pmatrix} \quad \text{span}\{u, w, z\} = \mathbb{R}^2$$

The vectors u , w and z span $\mathbb{R}^2$, as indicated by sketch below.

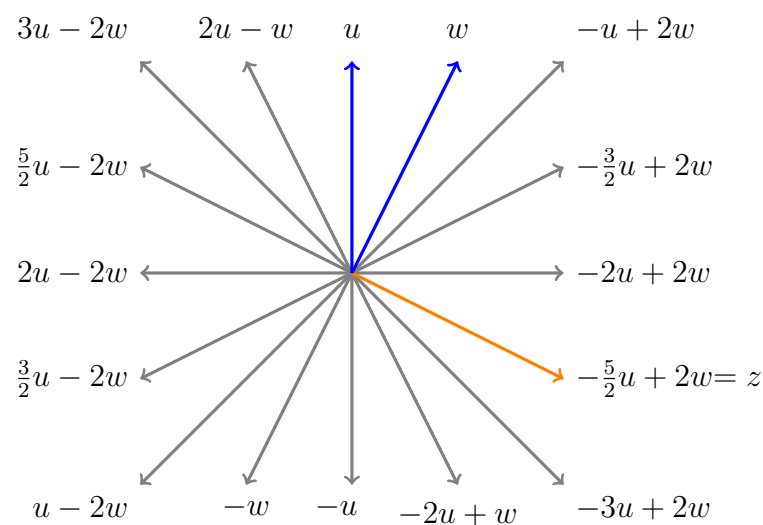
But we can leave out any one of u , w and z and still span $\mathbb{R}^2$!



38 Omitting redundant generators: picture 2 of 4

$$u = \begin{pmatrix} 0 \\ 2 \end{pmatrix} \quad w = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad z = \begin{pmatrix} 2 \\ -1 \end{pmatrix} \quad \text{span}\{u, w, z\} = \mathbb{R}^2$$

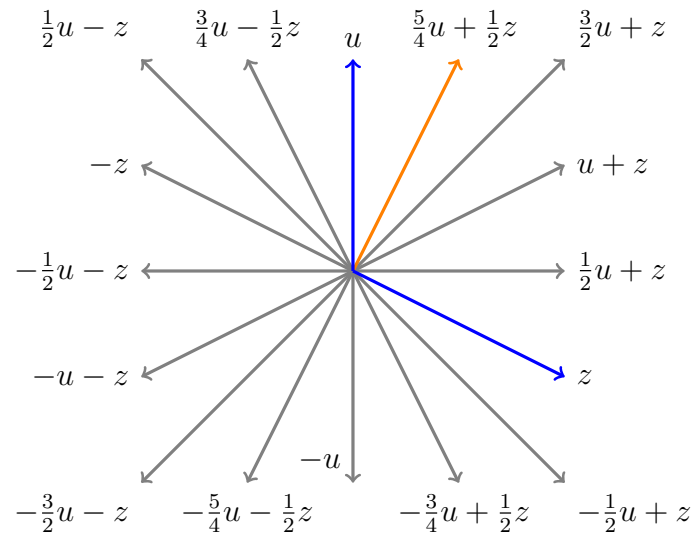
Omitting z : The vectors u and w alone already span $\mathbb{R}^2$, as indicated by sketch below: $z = -\frac{5}{2}u + 2w \in \text{span}\{u, w\} = \mathbb{R}^2$



39 Omitting redundant generators: picture 3 of 4

$$u = \begin{pmatrix} 0 \\ 2 \end{pmatrix} \quad w = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad z = \begin{pmatrix} 2 \\ -1 \end{pmatrix} \quad \text{span} \{u, w, z\} = \mathbb{R}^2$$

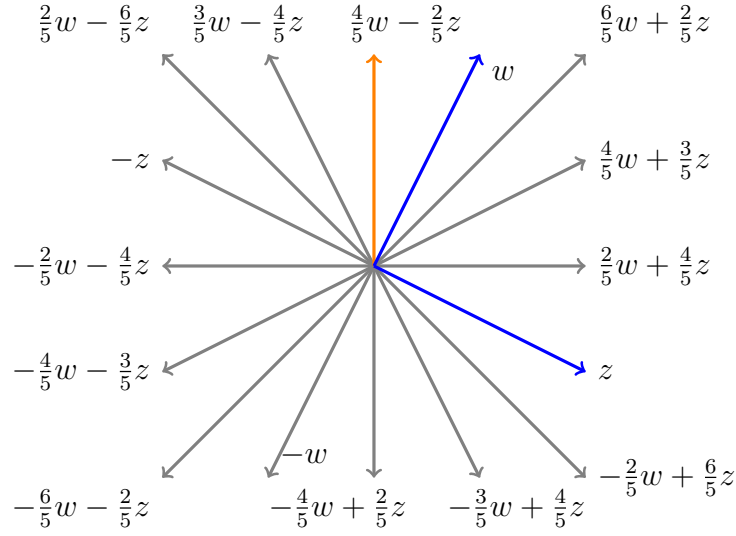
Omitting w : The vectors u and z alone already span $\mathbb{R}^2$, as indicated by sketch below: $w = \frac{5}{4}u + \frac{1}{2}z \in \text{span} \{u, z\} = \mathbb{R}^2$



40 Omitting redundant generators: picture 4 of 4

$$u = \begin{pmatrix} 0 \\ 2 \end{pmatrix} \quad w = \begin{pmatrix} 1 \\ 2 \end{pmatrix} \quad z = \begin{pmatrix} 2 \\ -1 \end{pmatrix} \quad \text{span} \{u, w, z\} = \mathbb{R}^2$$

Omitting u : The vectors w and z alone already span $\mathbb{R}^2$, as indicated by sketch below: $u = \frac{4}{5}w - \frac{2}{5}z \in \text{span} \{w, z\} = \mathbb{R}^2$



41 Example for omitting redundant generators

Consider the following sets of vectors in $\mathbb{R}^3$:

$$T = \left\{ \begin{pmatrix} 8 \\ 6 \\ 2 \end{pmatrix}, \begin{pmatrix} 4 \\ -2 \\ 6 \end{pmatrix} \right\} \text{ and } S = \left\{ \begin{pmatrix} 8 \\ 6 \\ 2 \end{pmatrix}, \begin{pmatrix} 4 \\ -2 \\ 6 \end{pmatrix}, \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \right\}$$

Claim: $\text{span}(S \setminus T) = \text{span}(S)$.

Proof. We have $T \subseteq \text{span}(S \setminus T) = \text{span}\left(\left\{ \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \right\}\right)$ because

$$\begin{pmatrix} 8 \\ 6 \\ 2 \end{pmatrix} = 5 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} + 3 \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix} \text{ and } \begin{pmatrix} 4 \\ -2 \\ 6 \end{pmatrix} = 5 \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} - \begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}.$$

So by “omitting redundant generators” (slide 36) we have the equality

$$\text{span}(S \setminus T) = \text{span}(S).$$

■

42 Another example for omitting redundant generators

$$v_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix} \quad v_2 = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} \quad w_1 = \begin{pmatrix} 4 \\ 2 \\ 4 \end{pmatrix} \quad w_2 = \begin{pmatrix} 0 \\ 4 \\ 0 \end{pmatrix}$$

Claim: $\text{span}\{v_1, v_2\} = \text{span}\{w_1, w_2\}$

Proof. $S = \{v_1, v_2, w_1, w_2\}$, $T = \{v_1, v_2\}$ and $Z = \{w_1, w_2\}$ so that $Z = S \setminus T$. Note $T \subseteq \text{span}(S \setminus T) = \text{span}(Z)$ since

$$v_1 = \frac{1}{4}w_1 - \frac{1}{8}w_2 \quad \text{and} \quad v_2 = \frac{1}{4}w_1 + \frac{3}{8}w_2 .$$

So we have $\text{span}(Z) = \text{span}(S \setminus T) = \text{span}(S)$.

Similarly, $Z \subseteq \text{span}(S \setminus Z) = \text{span}(T)$ as

$$w_1 = 3v_1 + v_2 \quad \text{and} \quad w_2 = -2v_1 + 2v_2 ,$$

and thus $\text{span}(T) = \text{span}(S \setminus Z) = \text{span}(S)$.

So both subspaces in the claim are equal to $\text{span}(S)$. ■

43 Summary of Chapter 3

- A *linear combination* of vectors $v_1, v_2, \dots, v_\ell$ is a vector v which can be written as $v = \lambda_1 v_1 + \lambda_2 v_2 + \dots + \lambda_\ell v_\ell$.
- A *subspace* of a vector space is a non-empty subset W such that addition of vectors in W , and multiplication of vectors in W by scalars results in elements of W .
- The *span* of a set of vectors S is the set $\text{span}(S)$ of all linear combinations of elements of S . The span is a subspace (of the vector space the set S lives in).
- “Omitting redundant generators” provides a means to make S smaller without changing $\text{span}(S)$.
- A *finitely generated* vector space V is one that can be written as $V = \text{span}(S)$, for a *finite* set S . Examples include $\mathbb{R}^n$, $\mathcal{P}_n$, $\mathcal{M}_{m,n}$ and the vector spaces $\mathcal{W}$ and $\mathcal{E}$ (see slides 12, 19, 22, 26 and 27 of Chapter 2).